TS. Please:
- Change all words underlined in BLUE to have initial cap
- Lowercase all words underlined in GREEN
- Number Examples, Theorems, etc. decimally by chapter in a single sequence, e.g. Example 3.1, Definition 3.2, Theorem 3.3, Remark 3.4 etc.
- Examples/Definitions/Claims/Remarks should be set in roman. Propositions/Theorems/Corollaries/Lemmas should remain italicised thoughout. Please amend.

~~Chapter~~ 5

Markov Decision Processes and Finite Horizon Dynamic Programming

In Chapter 3 we considered multi-stage decision problems for *deterministic* systems. In many problems of interest, the system dynamics also involves *randomness*, which leads us to stochastic decision problems. In this chapter we introduce the basic model of Markov Decision Processes (MDP), which will be considered throughout the rest of the book, and discuss optimal decision making in the finite horizon setting.

Chapter Orientation

Main Prerequisites:

- Deterministic decision processes (Chapter 3)
- Markov chains (Chapter 4)

Main terminology:

- $\mathtt{s}_\mathtt{t}$ state at time $\mathtt{t}$
- $\mathcal{S}_\mathtt{t}$ state space at time $\mathtt{t}$
- $\mathtt{a}_\mathtt{t}$ action at time $\mathtt{t}$

- $\mathcal{A}_{\mathtt{t}}(\mathtt{s})$ action space at time $\mathtt{t}$ for state $\mathtt{s}$
- $\mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a})$ transition probability at time $\mathtt{t}$
- $\mathtt{p}(\mathtt{s}'|\mathtt{s},\mathtt{a})$ stationary transition probability
- $\mathtt{h}_{\mathtt{t}}$ history at time $\mathtt{t}$
- π control policy
- $\pi_{\mathtt{t}}$ policy at time $\mathtt{t}$
- π^* optimal control policy
- Π_{HD},Π_{MD},Π_{SD} history-dependent, Markov~~,~~ and stationary deterministic policy sets
- Π_{HS},Π_{MS},Π_{SS} history-dependent, Markov~~,~~ and stationary stochastic policy sets
- $\mathtt{p}_0$ initial state distribution
- $\mathtt{r}_{\mathtt{t}}(\mathtt{s},\mathtt{a})$ reward function of state $\mathtt{s}$ and action $\mathtt{a}$ at time $\mathtt{t}$
- $\tilde{\mathtt{r}}_{\mathtt{t}}(\mathtt{s}_{\mathtt{t}},\mathtt{a}_{\mathtt{t}},\mathtt{s}_{\mathtt{t}+1})$ reward function of state $\mathtt{s}_{\mathtt{t}}$, action $\mathtt{a}_{\mathtt{t}}$, and next state $\mathtt{s}_{\mathtt{t}+1}$
- $R_{\mathtt{t}}$ reward obtained at time $\mathtt{t}$
- $\mathtt{R}_{\max}$ bound on reward
- $\mathtt{T}$ fixed time horizon
- $\mathcal{V}^{\pi}(\mathtt{s})$ expected cumulative reward starting from state $\mathtt{s}$ and following policy π for $\mathtt{T}$ steps
- $\mathcal{V}^{*}(\mathtt{s})$ maximal expected cumulative reward in $\mathtt{T}$ steps starting from state $\mathtt{s}$
- $\mathcal{V}^{\pi}_{\mathtt{t}}(\mathtt{s})$ value function: expected reward-to-go from state $\mathtt{s}$ at time $\mathtt{t}$ under policy π
- $\mathcal{V}^{*}_{\mathtt{t}}(\mathtt{s})$ optimal value function for state $\mathtt{s}$ and time $\mathtt{t}$

- $V_{\mathtt{t}}(\mathtt{s})$ value of state $\mathtt{s}$, time $\mathtt{t}$ in the dynamic programming algorithm
- $\mathcal{Q}^*_{\mathtt{t}}(\mathtt{s},\mathtt{a})$ state-action value function: maximal expected reward to go from state $\mathtt{s}$ and action $\mathtt{a}$ at time $\mathtt{t}$

5.1 Markov Decision Process

A Markov Decision Process consists of two main parts:

1. A controlled dynamic system, with stochastic evolution.
2. A performance objective to be optimized.

In this section we describe the first part, which is modeled as a controlled Markov chain.

Consider a controlled dynamic system, defined over:

- A discrete time axis $\mathbb{T} = \{0, 1, \ldots, \mathtt{T}-1\}$ (finite horizon), or $\mathbb{T} = \{0, 1, 2, \ldots\}$ (infinite horizon). To simplify the discussion we refer below to the infinite horizon case, which can always be "truncated" at $\mathtt{T}$ if needed.
- A finite state space $\mathcal{S}$, where $\mathcal{S}_{\mathtt{t}} \subset \mathcal{S}$ is the set of possible states at time $\mathtt{t} \in \mathbb{T}$.
- A finite action set $\mathcal{A}$, where $\mathcal{A}_{\mathtt{t}}(\mathtt{s}) \subset \mathcal{A}$ is the set of possible actions at time $\mathtt{t} \in \mathbb{T}$ and state $\mathtt{s} \in \mathcal{S}_{\mathtt{t}}$, and $\mathcal{A}_{\mathtt{t}} = \cup_{\mathtt{s}\in\mathcal{S}_{\mathtt{t}}} \mathcal{A}_{\mathtt{t}}(\mathtt{s})$ is the set of possible actions in time $\mathtt{t} \in \mathbb{T}$.

State transition probabilities: Suppose that at time $\mathtt{t}$ we are in state $\mathtt{s}_{\mathtt{t}} = \mathtt{s}$, and choose an action $\mathtt{a}_{\mathtt{t}} = \mathtt{a}$. The next state $\mathtt{s}_{\mathtt{t}+1} = \mathtt{s}'$ is then determined randomly according to a probability distribution $\mathtt{p}_{\mathtt{t}}(\cdot|\mathtt{s},\mathtt{a})$ on $\mathcal{S}_{\mathtt{t}+1}$. That is,

$$\Pr(\mathtt{s}_{\mathtt{t}+1} = \mathtt{s}'|\mathtt{s}_{\mathtt{t}} = \mathtt{s}, \mathtt{a}_{\mathtt{t}} = \mathtt{a}) = \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a}), \qquad \mathtt{s}' \in \mathcal{S}_{\mathtt{t}+1}$$

The probability $\mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a})$ is the *transition probability* from state $\mathtt{s}$ to state $\mathtt{s}'$ for a given action $\mathtt{a}$. We naturally require that $\mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a}) \geq 0$, and $\sum_{\mathtt{s}'\in\mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a}) = 1$ for all $\mathtt{s} \in \mathcal{S}_{\mathtt{t}}, \mathtt{a} \in \mathcal{A}_{\mathtt{t}}(\mathtt{s})$.

Implicit in this definition is the controlled-Markov property:

$$\Pr(\mathtt{s}_{\mathtt{t}+1} = \mathtt{s}'|\mathtt{s}_{\mathtt{t}}, \mathtt{a}_{\mathtt{t}}, \ldots, \mathtt{s}_{0,}\mathtt{a}_0) = \Pr(\mathtt{s}_{\mathtt{t}+1} = \mathtt{s}'|\mathtt{s}_{\mathtt{t}}, \mathtt{a}_{\mathtt{t}}).$$

The set of probability distributions

$$\mathtt{P} = \{\mathtt{p}_{\mathtt{t}}(\cdot|\mathtt{s},\mathtt{a}) \ : \ \mathtt{s} \in \mathcal{S}_{\mathtt{t}}, \mathtt{a} \in \mathcal{A}_{\mathtt{t}}(\mathtt{s}), \mathtt{t} \in \mathbb{T}\},$$

is called the *transition law* or *transition kernel* of the controlled Markov process.

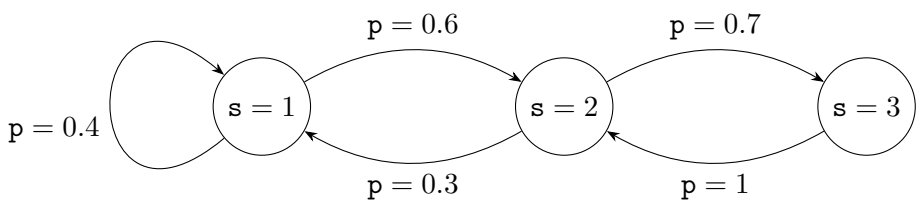

Figure 5.1: Markov chain

Stationary Models: The controlled Markov chain is called stationary or time-invariant if the transition probabilities do not depend on the time $\mathtt{t}$. That is:

$$\forall \mathtt{t}, \quad \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a}) \equiv \mathtt{p}(\mathtt{s}'|\mathtt{s},\mathtt{a}), \;\; \mathcal{S}_{\mathtt{t}} \equiv \mathcal{S}, \;\; \mathcal{A}_{\mathtt{t}}(\mathtt{s}) \equiv \mathcal{A}(\mathtt{s}).$$

Graphical Notation: The state transition probabilities of a Markov chain are often illustrated via a state transition diagram, such as in Figure 5.1 and 5.2.

In figure 5.1 we have three states $\mathcal{S} = \{1, 2, 3\}$. The transition are marked by the arrows, and labeled with their probabilities. For example: from state 1 we transition to state 2 with probability 0.6 and with probability 0.4 we stay in state 1.

A graphical description of a controlled Markov chain is a bit more complicated because of the additional action variable. We obtain the diagram (drawn for state $\mathtt{s} = 1$ only, and for a given time $\mathtt{t}$) in Figure 5.2, reflecting the following transition probabilities:

$$\mathtt{p}(\mathtt{s}' = 2|\mathtt{s} = 1, \mathtt{a} = 1) = 1$$
$$\mathtt{p}(\mathtt{s}'|\mathtt{s} = 1, \mathtt{a} = 2) = \begin{cases} 0.3 & : \; \mathtt{s}' = 1 \\ 0.2 & : \; \mathtt{s}' = 2 \\ 0.5 & : \; \mathtt{s}' = 3 \end{cases}$$

State-equation notation: The stochastic state dynamics can be equivalently defined in terms of a state equation of the form

$$\mathtt{s}_{\mathtt{t}+1} = f_{\mathtt{t}}(\mathtt{s}_{\mathtt{t}}, \mathtt{a}_{\mathtt{t}}, w_{\mathtt{t}}),$$

where $w_{\mathtt{t}}$ is a random variable (RV). If $(w_{\mathtt{t}})_{\mathtt{t}\geq 0}$ is a sequence of independent RVs, and further each $w_{\mathtt{t}}$ is independent of the "past" $(\mathtt{s}_{\mathtt{t}-1}, \mathtt{a}_{\mathtt{t}-1}, \ldots \mathtt{s}_0)$, then $(\mathtt{s}_{\mathtt{t}}, \mathtt{a}_{\mathtt{t}})_{\mathtt{t}\geq 0}$ is a controlled Markov process. For example, the state transition law of the last

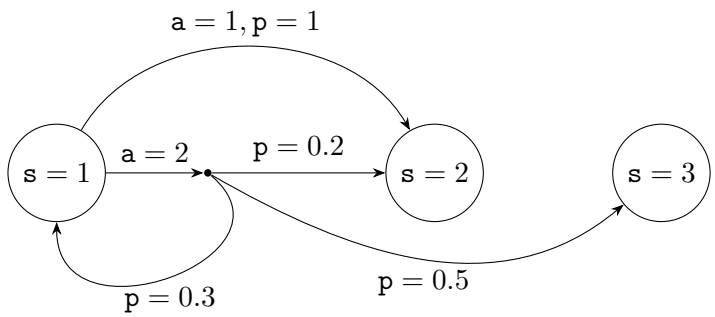

Figure 5.2~~:~~ Controlled Markov chain

example can be written in this way, using $w_{\mathtt{t}} \in \{4, 5, 6\}$, with $p_w(4) = 0.3,\ p_w(5) = 0.2,\ p_w(6) = 0.5$ and, for $\mathtt{s}_{\mathtt{t}} = 1$:

$$\begin{aligned} f_{\mathtt{t}}(1, 1, w_{\mathtt{t}}) &= 2 \\ f_{\mathtt{t}}(1, 2, w_{\mathtt{t}}) &= w_{\mathtt{t}} - 3 \end{aligned}$$

This state algebraic equation notation is especially useful for problems with continuous state space~~,~~ but also for some models with discrete states. Equivalently, we can write

$$f_{\mathtt{t}}(1, 2, w_{\mathtt{t}}) = 1 \cdot \mathbb{I}[w_{\mathtt{t}} = 4] + 2 \cdot \mathbb{I}[w_{\mathtt{t}} = 5] + 3 \cdot \mathbb{I}[w_{\mathtt{t}} = 6],$$

where $\mathbb{I}[\cdot]$ is the indicator function.

Next we recall the definitions of control policies from Chapter 3.

Control Policies

- A general or **history-dependent deterministic** control policy $\pi = (\pi_{\mathtt{t}})_{\mathtt{t} \in \mathbb{T}}$ is a mapping from each possible history $\mathtt{h}_{\mathtt{t}} = (\mathtt{s}_0, \mathtt{a}_0, \ldots, \mathtt{s}_{\mathtt{t}-1}, \mathtt{a}_{\mathtt{t}-1}, \mathtt{s}_{\mathtt{t}})$, and time $\mathtt{t} \in \mathbb{T}$, to an action $\mathtt{a}_{\mathtt{t}} = \pi_{\mathtt{t}}(\mathtt{h}_{\mathtt{t}}) \in \mathcal{A}_{\mathtt{t}}$. We denote the set of general policies by Π_{HD}.

- A **Markov deterministic** control policy π is allowed to depend on the current state and time only, i.e., $\mathtt{a}_{\mathtt{t}} = \pi_{\mathtt{t}}(\mathtt{s}_{\mathtt{t}})$. We denote the set of Markov deterministic policies by Π_{MD}.

- For stationary models, we may define **stationary deterministic** control policies that depend on the current state alone. A stationary policy is defined by a single mapping $\pi : \mathcal{S} \to \mathcal{A}$, so that $\mathtt{a}_{\mathtt{t}} = \pi(\mathtt{s}_{\mathtt{t}})$ for all $\mathtt{t} \in \mathbb{T}$. We denote the set of stationary policies by Π_{SD}.

- Evidently, $\Pi_{HD} \supset \Pi_{MD} \supset \Pi_{SD}$.

Randomized Control policies

- The control policies defined above specify deterministically the action to be taken at each stage. In some cases we want to allow for a random choice of action.

- A general randomized control policy assigns to each possible history $\mathtt{h}_\mathtt{t}$ a probability distribution $\pi_\mathtt{t}(\cdot|\mathtt{h}_\mathtt{t})$ over the action set $\mathcal{A}_\mathtt{t}$. That is, $\Pr(\mathtt{a}_\mathtt{t} = \mathtt{a}|\mathtt{h}_\mathtt{t}) = \pi_\mathtt{t}(\mathtt{a}|\mathtt{h}_\mathtt{t})$. We denote the set of history-dependent stochastic policies by Π_{HS}.

- Similarly, we can define the set Π_{MS} of Markov stochastic control policies, where $\pi_\mathtt{t}(\cdot|\mathtt{h}_\mathtt{t})$ is replaced by $\pi_\mathtt{t}(\cdot|\mathtt{s}_\mathtt{t})$, and the set Π_{SS} of stationary stochastic control policies, where $\pi_\mathtt{t}(\cdot|\mathtt{s}_\mathtt{t})$ is replaced by $\pi(\cdot|\mathtt{s}_\mathtt{t})$, namely the policy is independent of the time.

- Note that the set Π_{HS} includes all other policy sets as special cases.

The Induced Stochastic Process Let $\mathtt{p}_0 = \{\mathtt{p}_0(\mathtt{s}), \mathtt{s} \in \mathcal{S}_0\}$ be a probability distribution for the initial state $\mathtt{s}_0$. (Many times we will assume that the initial state is deterministic and given by $\mathtt{s}_0$.) A control policy $\pi \in \Pi_{HS}$, together with the transition law $\mathtt{P} = \{\mathtt{p}_\mathtt{t}(\mathtt{s}'|\mathtt{s},\mathtt{a})\}$ and the initial state distribution $\mathtt{p}_0 = (\mathtt{p}_0(\mathtt{s}),\ \mathtt{s} \in \mathcal{S}_0)$, induces a probability distribution over any finite state-action sequence $\mathtt{h}_\mathtt{T} = (\mathtt{s}_0, \mathtt{a}_0, \ldots, \mathtt{s}_{\mathtt{T}-1}, \mathtt{a}_{\mathtt{T}-1}, \mathtt{s}_\mathtt{T})$, given by

$$\Pr(\mathtt{h}_\mathtt{T}) = \mathtt{p}_0(\mathtt{s}_0) \prod_{\mathtt{t}=0}^{\mathtt{T}-1} \mathtt{p}_\mathtt{t}(\mathtt{s}_{\mathtt{t}+1}|\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t}) \pi_\mathtt{t}(\mathtt{a}_\mathtt{t}|\mathtt{h}_\mathtt{t}),$$

where $\mathtt{h}_\mathtt{t} = (\mathtt{s}_0, \mathtt{a}_0, \ldots, \mathtt{s}_{\mathtt{T}-1}, \mathtt{a}_{\mathtt{T}-1}, \mathtt{s}_\mathtt{t})$. To see this, observe the recursive relation:

$$\begin{aligned}\Pr(\mathtt{h}_{\mathtt{t}+1}) &= \Pr(\mathtt{h}_\mathtt{t}, \mathtt{a}_\mathtt{t}, \mathtt{s}_{\mathtt{t}+1}) = \Pr(\mathtt{s}_{\mathtt{t}+1}|\mathtt{h}_\mathtt{t}, \mathtt{a}_\mathtt{t}) \Pr(\mathtt{a}_\mathtt{t}|\mathtt{h}_\mathtt{t}) \Pr(\mathtt{h}_\mathtt{t}) \\ &= \mathtt{p}_\mathtt{t}(\mathtt{s}_{\mathtt{t}+1}|\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t}) \pi_\mathtt{t}(\mathtt{a}_\mathtt{t}|\mathtt{h}_\mathtt{t}) \Pr(\mathtt{h}_\mathtt{t}).\end{aligned}$$

In the last step we used the conditional Markov property of the controlled chain: $\Pr(\mathtt{s}_{\mathtt{t}+1}|\mathtt{h}_\mathtt{t}, \mathtt{a}_\mathtt{t}) = \mathtt{p}_\mathtt{t}(\mathtt{s}_{\mathtt{t}+1}|\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t})$, and the definition of the control policy $\pi_\mathtt{t}$. The required formula follows by recursion.

Therefore, the state-action sequence $\mathtt{h}_\infty = (\mathtt{s}_k, \mathtt{a}_k)_{k \geq 0}$ can now be considered a stochastic process. We denote the probability law of this stochastic process by

$\Pr^{\pi,\mathtt{p}_0}(\cdot)$. The corresponding expectation operator is denoted by $\mathbb{E}^{\pi,\mathtt{p}_0}(\cdot)$, where the expectation is over state-action sequences generated using the stochastic process $\Pr^{\pi,\mathtt{p}_0}(\cdot)$. When the initial state $\mathtt{s}_0$ is deterministic (i.e., $\mathtt{p}_0(\mathtt{s})$ is concentrated on a single state $\mathtt{s}$), we may simply write $\Pr^{\pi,s}(\cdot)$ or $\Pr^{\pi}(\cdot|\mathtt{s}_0 = \mathtt{s})$.

Under a Markov control policy, the state sequence $(\mathtt{s}_\mathtt{t})_{\mathtt{t}\geq 0}$ becomes a *Markov chain*, with transition probabilities~~:~~

$$\Pr(\mathtt{s}_{\mathtt{t}+1} = \mathtt{s}'|\mathtt{s}_\mathtt{t} = \mathtt{s}) = \sum_{\mathtt{a}\in\mathcal{A}_\mathtt{t}} \mathtt{p}_\mathtt{t}(\mathtt{s}'|\mathtt{s},\mathtt{a})\pi_\mathtt{t}(\mathtt{a}|\mathtt{s}).$$

This follows since~~:~~

$$\begin{aligned}
\Pr(\mathtt{s}_{\mathtt{t}+1} = \mathtt{s}'|\mathtt{s}_\mathtt{t} = \mathtt{s}) &= \sum_{\mathtt{a}\in\mathcal{A}_\mathtt{t}} \Pr(\mathtt{s}_{\mathtt{t}+1} = \mathtt{s}', \mathtt{a}_\mathtt{t} = \mathtt{a}|\mathtt{s}_\mathtt{t} = \mathtt{s}) \\
&= \sum_{\mathtt{a}\in\mathcal{A}_\mathtt{t}} \Pr(\mathtt{s}_{\mathtt{t}+1} = \mathtt{s}'|\mathtt{s}_\mathtt{t} = \mathtt{s}, \mathtt{a}_\mathtt{t} = \mathtt{a}) \Pr(\mathtt{a}_\mathtt{t} = \mathtt{a}|\mathtt{s}_\mathtt{t} = \mathtt{s}) \\
&= \sum_{\mathtt{a}\in\mathcal{A}_\mathtt{t}} \mathtt{p}_\mathtt{t}(\mathtt{s}'|\mathtt{s},\mathtt{a})\pi_\mathtt{t}(\mathtt{a}|\mathtt{s}).
\end{aligned}$$

If the controlled Markov chain is stationary (time-invariant) and the control policy is stationary, then the induced Markov chain is stationary as well.

Remark 5.1. *For most non-learning optimization problems, Markov policies suffice to achieve the optimum.*

Remark 5.2. *Implicit in these definitions of control policies is the assumption that the current state $\mathtt{s}_\mathtt{t}$ can be fully observed before the action $\mathtt{a}_\mathtt{t}$ is chosen . If this is not the case we need to consider the problem of a Partially Observed MDP (POMDP), which is more involved and is not discussed in this book.*

5.2 Performance Criteria

5.2.1 Finite Horizon Return

Consider the finite-horizon return, with a fixed time horizon $\mathtt{T}$. As in the deterministic case, we are given an immediate reward function $\mathtt{r}_\mathtt{t} = \{\mathtt{r}_\mathtt{t}(\mathtt{s},\mathtt{a}) : \mathtt{s} \in \mathcal{S}_\mathtt{t}, \mathtt{a} \in \mathcal{A}_\mathtt{t}\}$ for $0 \leq \mathtt{t} \leq \mathtt{T}-1$, and a terminal reward function $\mathtt{r}_\mathtt{T} = \{\mathtt{r}_\mathtt{T}(\mathtt{s}) : \mathtt{s} \in \mathcal{S}_\mathtt{T}\}$. We assume that reward functions are bounded, $|\mathtt{r}_\mathtt{t}|, |\mathtt{r}_\mathtt{T}| \in [0, \mathtt{R}_{\max}]$. The obtained reward is $R_\mathtt{t} = \mathtt{r}_\mathtt{t}(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t})$ at times $\mathtt{t} \leq \mathtt{T}-1$, and $R_\mathtt{T} = \mathtt{r}_\mathtt{T}(\mathtt{s}_\mathtt{T})$ at the last stage. (Note

that $\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t}$ and $\mathtt{s}_\mathtt{T}$ are random variables that depend both on the policy π and the stochastic transitions.) Our general goal is to maximize the cumulative return:

$$\sum_{\mathtt{t}=0}^{\mathtt{T}} R_\mathtt{t} = \sum_{\mathtt{t}=0}^{\mathtt{T}-1} \mathtt{r}_\mathtt{t}(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t}) + \mathtt{r}_\mathtt{T}(\mathtt{s}_\mathtt{T}).$$

However, since the system is stochastic, the cumulative return will generally be a random variable, and we need to specify in which sense to maximize it. A natural first option is to consider the expected value of the return. That is, define:

$$\mathcal{V}^\pi(\mathtt{s}) = \mathbb{E}^\pi\Big[\sum_{\mathtt{t}=0}^{\mathtt{T}} R_\mathtt{t} | \mathtt{s}_0 = \mathtt{s}\Big] \equiv \mathbb{E}^{\pi,\mathtt{s}}\Big[\sum_{\mathtt{t}=0}^{\mathtt{T}} R_\mathtt{t}\Big].$$

Here π is the control policy as defined above, and $\mathtt{s}$ denotes the initial state. Hence, $\mathcal{V}^\pi(\mathtt{s})$ is the expected cumulative return under the control policy π. Our goal is to find an optimal control policy that maximizes $\mathcal{V}^\pi(\mathtt{s})$.

Remark 5.3. *Reward dependence on the next state: In some problems, the obtained reward may depend on the next state as well:* $R_\mathtt{t} = \tilde{\mathtt{r}}_\mathtt{t}(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t}, \mathtt{s}_{\mathtt{t}+1})$. *For control purposes, when we only consider the expected value of the reward, we can reduce this reward function to the usual one by defining*

$$\mathtt{r}_\mathtt{t}(\mathtt{s}, \mathtt{a}) \triangleq \mathbb{E}[R_\mathtt{t} | \mathtt{s}_\mathtt{t} = \mathtt{s}, \mathtt{a}_\mathtt{t} = \mathtt{a}] \equiv \sum_{\mathtt{s}' \in S} \mathtt{p}(\mathtt{s}'|\mathtt{s}, \mathtt{a}) \tilde{\mathtt{r}}_\mathtt{t}(\mathtt{s}, \mathtt{a}, \mathtt{s}').$$

Remark 5.4. *Random rewards: The reward* $R_\mathtt{t}$ *may also be random, namely a random variable whose distribution depends on* $(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t})$. *This can also be reduced to our standard model for planning purposes by looking at the expected value of* $R_\mathtt{t}$, *namely*

$$\mathtt{r}_\mathtt{t}(\mathtt{s}, \mathtt{a}) = \mathbb{E}[R_\mathtt{t} | \mathtt{s}_\mathtt{t} = \mathtt{s}, \mathtt{a}_\mathtt{t} = \mathtt{a}].$$

Remark 5.5. *Risk-sensitive criteria: The expected cumulative return is by far the most common goal for planning. However, it is not the only one possible. For example, one may consider the following risk-sensitive return function:*

$$\mathcal{V}^\pi_\lambda(\mathtt{s}) = \frac{1}{\lambda} \log \mathbb{E}^{\pi,\mathtt{s}} \left(\exp\big(\lambda \sum_{\mathtt{t}=0}^{\mathtt{T}} R_\mathtt{t}\big) \right).$$

For $\lambda > 0$, *the exponent gives higher weight to high rewards, and the opposite for* $\lambda < 0$.

The key challenge: stochastic transitions It is important to understand what is the 'key' challenge in MDPs – is it the stochastic rewards, the stochastic transitions, or both. The next several results show that an MDP with stochastic rewards may replaced with an MDP with deterministic rewards. The opposite is not true.

In the case that the rewards are stochastic, but have a discrete support, we can construct an equivalent MDP in which all the rewards are deterministic and trajectories have the same distribution of rewards. This implies that the important challenge is the stochastic state transition function, and the rewards can be assumed to be deterministic. Formally, given a trajectory we define a *rewards trajectory* as the sub-trajectory that includes only the rewards, i.e., for a trajectory $(\mathtt{s}_0, \mathtt{a}_0, \mathtt{r}_0, \mathtt{s}_1, \ldots)$ the reward trajectory is $(\mathtt{r}_0, \mathtt{r}_1, \ldots)$.

Theorem 5.1. *Given an MDP $M(\mathcal{S}, \mathcal{A}, \mathtt{P}, \mathtt{r}, \mathtt{s}_0)$, where the rewards are stochastic, with support $\mathcal{K} = \{0, \ldots, k-1\}$, there is an MDP $M'(\mathcal{S} \times \mathcal{K}, \mathcal{A}, \mathtt{P}', \mathtt{r}', \mathtt{s}'_0)$, where the rewards are deterministic, and a mapping of policies π of M to π' policies of M', such that: running π in M for horizon $\mathtt{T}$ generates reward trajectory $R = (R_0, \ldots, R_\mathtt{T})$, and running π' in M' for horizon $\mathtt{T}+1$ generates reward trajectory $R' = (R_1, \ldots, R_{\mathtt{T}+1})$, and the distributions of R and R' are identical.*

Proof. For simplicity we assume that the MDP is loop-free, namely you can reach any state at most once in a trajectory. This is mainly to simplify the notation.

The basic idea is to encode the rewards in the states of M', which are $\mathcal{S} \times \mathcal{K} = \mathcal{S}'$. For each $(\mathtt{s}, i) \in \mathcal{S}'$ and action $\mathtt{a} \in \mathcal{A}$ we have $\mathtt{p}'_\mathtt{t}((\mathtt{s}', j)|(\mathtt{s}, i), \mathtt{a}) = \mathtt{p}_\mathtt{t}(\mathtt{s}'|\mathtt{s}, \mathtt{a}) \Pr[R_\mathtt{t}(\mathtt{s}, \mathtt{a}) = j]$, and $\mathtt{p}'_\mathtt{T}((\mathtt{s}', j)|(\mathtt{s}, i)) = \mathbb{I}(\mathtt{s}' = \mathtt{s}) \Pr[R_\mathtt{T}(\mathtt{s}) = j]$. The reward is $\mathtt{r}'_\mathtt{t}((\mathtt{s}, i), \mathtt{a}) = i$. The initial state is $\mathtt{s}'_0 = (\mathtt{s}_0, 0)$.

For any policy $\pi(\mathtt{a}|\mathtt{s})$ in M we have a policy π' in M' where $\pi'(\mathtt{a}|(\mathtt{s}, i)) = \pi(\mathtt{a}|\mathtt{s})$.

We map trajectories of M to trajectories of M' which have identical probabilities. A trajectory $(\mathtt{s}_0, \mathtt{a}_0, R_0, \mathtt{s}_1, \mathtt{a}_1, R_1, \mathtt{s}_2 \ldots, R_\mathtt{T})$ is mapped to $((\mathtt{s}_0, 0), \mathtt{a}_0, 0, (\mathtt{s}_1, R_0), \mathtt{a}_1, R_0, (\mathtt{s}_2, R_1) \ldots, R_{\mathtt{T}+1})$. Let R and R' be the respective reward trajectories. Clearly, the two trajectories have identical probabilities. This implies that the rewards trajectories R and R' have identical probabilities (up to a shift of one in the index). □

Theorem 5.1 requires the number of rewards to be bounded, and guarantees that the reward distribution be identical. In the case that the rewards are continuous, we can have a similar guarantee for expected return functions.

Theorem 5.2. *Given an MDP $M(\mathcal{S}, \mathcal{A}, \mathtt{P}, \mathtt{r}, \mathtt{s}_0)$, where the rewards are stochastic, with support $[0, 1]$, there is an MDP $M'(\mathcal{S}, \mathcal{A}, \mathtt{P}, \mathtt{r}', \mathtt{s}_0)$, where the rewards are stochastic, with support $\{0, 1\}$, such that for any policy $\pi \in \Pi_{MS}$ the distribution of the expected rewards trajectory is identical.*

Proof. We simply change the reward of $(\mathtt{s}, \mathtt{a})$ to be $\{0, 1\}$ by changing them to be ~~a~~ Bernoulli random variables with ~~a~~ parameter $\mathtt{r_t}(\mathtt{s}, \mathtt{a})$, i.e., $\Pr[R_\mathtt{t} = 1|\mathtt{s}, \mathtt{a}] = \mathtt{r_t}(\mathtt{s}, \mathtt{a})$ and $\Pr[R_\mathtt{t} = 0|\mathtt{s}, \mathtt{a}] = 1 - \mathtt{r_t}(\mathtt{s}, \mathtt{a})$, and we recall that $\mathtt{r_t}(\mathtt{s}, \mathtt{a})$ denotes the expected reward for $\mathtt{s}, \mathtt{a}$ in the original MDP M. Clearly, the expected value of the rewards in M and M' is identical. Further, since $\pi \in \Pi_{MS}$, it depends only on $\mathtt{s}$ and $\mathtt{t}$, which implies that the behavior (states and actions) will be identical in M and M'. □

We have also established the following corollary.

Corollary 5.3. *Given an MDP $M(\mathcal{S}, \mathcal{A}, \mathtt{P}, \mathtt{r}, \mathtt{s}_0)$ and horizon $\mathtt{T}$, where the rewards are stochastic with support $[0, 1]$, there is an MDP $M'(\mathcal{S} \times \{0, 1\}, \mathcal{A}, \mathtt{P}', \mathtt{r}', \mathtt{s}'_0)$, and horizon $\mathtt{T}+1$, where the rewards are deterministic, and a mapping of policies $\pi \in \Pi_{MS}$ of M to $\pi' \in \Pi_{MD}$ policies of M', such that $\mathcal{V}^{\pi,M}(\mathtt{s}_0) = \mathcal{V}^{\pi',M'}(\mathtt{s}'_0)$.*

5.2.2 Infinite Horizon Problems

We next consider planning problems that extend to an infinite time horizon, $\mathtt{t} = 0, 1, 2, \ldots$~~.~~ Such planning problems arise when the system in question is expected to operate for a long time, or a large number of steps, possibly with no specific "closing" time. Infinite horizon problems are most often defined for stationary problems. In that case, they enjoy the important advantage that optimal policies can be found among the class of stationary policies. We will restrict attention here to stationary models. As before, we have the immediate reward function $\mathtt{r}(\mathtt{s}, \mathtt{a})$, which extends to all $\mathtt{t} \geq 0$. The expected reward obtained at stage $\mathtt{t}$ is $\mathbb{E}[R_\mathtt{t}] = \mathtt{r}(\mathtt{s_t}, \mathtt{a_t})$.

Discounted return: The most common performance criterion for infinite horizon problems is the expected discounted return:

$$\mathcal{V}^\pi_\gamma(\mathtt{s}) = \mathbb{E}^\pi\Big[\sum_{\mathtt{t}=0}^{\infty} \gamma^t \mathtt{r}(\mathtt{s_t}, \mathtt{a_t})\Big|\mathtt{s}_0 = \mathtt{s}\Big] \equiv \mathbb{E}^{\pi,s}\Big[\sum_{\mathtt{t}=0}^{\infty} \gamma^t \mathtt{r}(\mathtt{s_t}, \mathtt{a_t})\Big],$$

where $0 < \gamma < 1$ is the discount factor. Mathematically, the discount factor ensures convergence of the sum (whenever the reward sequence is unbounded). This makes the problem "well behaved"~~,~~ and relatively easy to analyze. The discounted return is discussed in Chapter 6.

Average return: Here we are interested to maximize the long-term average return. The most common definition of the long-term average return is~~,~~

$$\mathcal{V}^\pi_{avg}(\mathtt{s}) = \liminf_{\mathtt{T}\to\infty} \mathbb{E}^{\pi,s}\left[\frac{1}{\mathtt{T}}\sum_{\mathtt{t}=0}^{\mathtt{T}-1} \mathtt{r}(\mathtt{s_t}, \mathtt{a_t})\right].$$

The theory of average-return planning problems is more involved~~,~~ and relies to a larger extent on the theory of Markov chains. (See Chapter 4 for planning in Deterministic Decision Processes with the average return criteria.)

5.2.3 Stochastic Shortest-Path Problems

In an important class of planning problems, the time horizon is not set beforehand, but rather the problem continues until a certain event occurs. This event can be defined as reaching some goal state. Let $\mathcal{S}_G \subset \mathcal{S}$ define the set of *goal states*. Define

$$\tau = \inf\{\mathtt{t} \geq 0 : \mathtt{s}_\mathtt{t} \in \mathcal{S}_G\}$$

as the first time in which a goal state is reached. The total expected return for this problem is defined as~~:~~

$$\mathcal{V}^\pi_{ssp}(\mathtt{s}) = \mathbb{E}^{\pi,\mathtt{s}}\left[\sum_{\mathtt{t}=0}^{\tau-1} \mathtt{r}(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t}) + \mathtt{r}_G(\mathtt{s}_\tau)\right].$$

Here $\mathtt{r}_G(\mathtt{s})$, $\mathtt{s} \in \mathcal{S}_G$ specified the reward at goal states. Note that the length of the run, τ, is a random variable.

Stochastic shortest path includes, naturally, the finite horizon case. This can be shown by creating a leveled MDP where at each time step we move to the next level and terminate at level $\mathtt{T}$. Specifically, we define a new state space $\mathcal{S}' = \mathcal{S} \times \mathbb{T}$, transition function $\mathtt{p}((\mathtt{s}', \mathtt{t}+1)|(\mathtt{s}, \mathtt{t}), \mathtt{a}) = \mathtt{p}(\mathtt{s}'|\mathtt{s}, \mathtt{a})$ and goal states $\mathcal{S}_G = \{(\mathtt{s}, \mathtt{T}) : \mathtt{s} \in \mathcal{S}\}$.

Stochastic shortest path includes also the discounted infinite horizon. To see that, add a new goal state, and from each state with probability $1-\gamma$ jump to the goal state and terminate. The expected return of a policy would be the same in both models. Specifically, we add a state $\mathtt{s}_G$ and modify the transition probability to $\mathtt{p}'$, such that $\mathtt{p}'(\mathtt{s}_G|\mathtt{s}, \mathtt{a}) = 1-\gamma$, for any state $\mathtt{s} \in \mathcal{S}$ and action $\mathtt{a} \in \mathcal{A}$ and $\mathtt{p}'(\mathtt{s}'|\mathtt{s}, \mathtt{a}) = \gamma \mathtt{p}(\mathtt{s}'|\mathtt{s}, \mathtt{a})$. The probability that we do not terminate by time $\mathtt{t}$ is exactly $\gamma^\mathtt{t}$. Therefore, the expected return is $\mathbb{E}^{\pi,s}[\sum_{\mathtt{t}=0}^{\infty} \gamma^t \mathtt{r}(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t})]$, which is identical to the discounted return.

This class of problems provides a natural extension of the standard shortest-path problem to stochastic settings. Some conditions on the system dynamics and reward function must be imposed for the problem to be well posed (e.g., that a goal state may be reached with probability one). Stochastic shortest path problems are also known as *episodic* MDP problems. (See ~~more in~~ Chapter 7.)

5.3 Sufficiency of Markov Policies

In all the performance criteria defined above, the criterion is composed of sums of terms of the form $\mathbb{E}[\mathtt{r}_\mathtt{t}(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t})]$. It follows that if two control policies induce the same marginal probability distributions $q_\mathtt{t}(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t})$ over the state-action pairs $(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t})$ for all $\mathtt{t} \geq 0$, they will have the same performance for any linear return function.

Using this observation, the next claim implies that it is enough to consider the set of (stochastic) Markov policies in the above planning problems.

Proposition 5.4. *Let $\pi \in \Pi_{HS}$ be a general (history-dependent, stochastic) control policy. Let*

$$p_\mathtt{t}^{\pi,\mathtt{s}_0}(\mathtt{s}, \mathtt{a}) = P^{\pi,\mathtt{s}_0}(\mathtt{s}_\mathtt{t} = \mathtt{s}, \mathtt{a}_\mathtt{t} = \mathtt{a}), \qquad (\mathtt{s}, \mathtt{a}) \in \mathcal{S}_\mathtt{t} \times \mathcal{A}_\mathtt{t}$$

Denote the marginal distributions induced by $q_\mathtt{t}(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t})$ on the state-action pairs $(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t})$, for all $\mathtt{t} \geq 0$. Then there exists a stochastic Markov policy $\tilde{\pi} \in \Pi_{MS}$ that induces the same marginal probabilities (for all initial states $\mathtt{s}_0$).

In Chapter 3 we showed for Deterministic Decision Process for the finite horizon that there is an optimal deterministic policy. The proof that every stochastic history dependent strategy has an equivalent stochastic Markovian policy (Theorem 3.1) showed how to generate the same state-action distribution, and applies to other setting as well. The proof that every stochastic Markovian policy has an equivalent (or better) deterministic Markovian policy (Theorem 3.2) depended on the finite horizon, but it is easy to extend it to any linear return function as well. (We leave the formal proof as an exercise to the reader.)

5.4 Finite-Horizon Dynamic Programming

Recall that we consider the expected total reward criterion, which we denote as

$$\mathcal{V}^\pi(\mathtt{s}_0) = \mathbb{E}^{\pi,\mathtt{s}_0}\left[\sum_{\mathtt{t}=0}^{\mathtt{T}-1} \mathtt{r}_\mathtt{t}(\mathtt{s}_\mathtt{t}, \mathtt{a}_\mathtt{t}) + \mathtt{r}_\mathtt{T}(\mathtt{s}_\mathtt{T})\right],$$

where π is the control policy used, and $\mathtt{s}_0$ is a given initial state. We wish to maximize the expected return $\mathcal{V}^\pi(\mathtt{s}_0)$ over all control policies, and find an optimal policy π^* that achieves the maximal expected return $\mathcal{V}^*(\mathtt{s}_0)$ for all initial states $\mathtt{s}_0$. Thus,

$$\mathcal{V}^*(\mathtt{s}_0) \triangleq \mathcal{V}^{\pi^*}(\mathtt{s}_0) = \max_{\pi \in \Pi_{HS}} \mathcal{V}^\pi(\mathtt{s}_0).$$

5.4.1 The Principle of Optimality

The celebrated principle of optimality (stated by Bellman) applies to a large class of multi-stage optimization problems~~,~~ and is at the heart of Dynamic Programming. As a general principle, it states that:

The tail of an optimal policy is optimal for the "tail problem".

This principle is not an actual claim, but rather a guiding principle that can be applied in different ways to each problem. For example, considering our finite-horizon problem, let $\pi^* = (\pi_0, \ldots, \pi_{\mathtt{T}-1})$ denote an optimal Markov policy, where $\pi_{\mathtt{t}}$ is the policy used at time $\mathtt{t}$ in the horizon. Take any state $\mathtt{s}_{\mathtt{t}} = \mathtt{s}'$ which has a positive probability to be reached under π^*, namely $P^{\pi^*,\mathtt{s}_0}(\mathtt{s}_{\mathtt{t}} = \mathtt{s}') > 0$. Then the tail policy $\pi^*_{\mathtt{t}:\mathtt{T}} = (\pi_{\mathtt{t}}, \ldots, \pi_{\mathtt{T}-1})$ is optimal for the "tail" criterion $\mathcal{V}^{\pi}_{\mathtt{t}:\mathtt{T}}(\mathtt{s}') = \mathbb{E}^{\pi}\left[\sum_{k=\mathtt{t}}^{\mathtt{T}} R_k | \mathtt{s}_{\mathtt{t}} = \mathtt{s}'\right]$.

Note that the reverse is not true. The prefix of the optimal policy is not optimal for the "prefix" problem. When we plan for a long horizon, we might start with non-greedy actions, so we can improve our return in later time steps. Specifically, the first action taken does not have to be the optimal action for horizon $\mathtt{T} = 1$, for which the greedy action is optimal.

5.4.2 Dynamic Programming for Policy Evaluation

As a "warmup", let us evaluate the reward of a given policy. Let $\pi = (\pi_0, \ldots, \pi_{\mathtt{T}-1})$ be a given deterministic Markov policy. Define the following reward-to-go function, or *value* function:

$$\mathcal{V}^{\pi}_{\mathtt{t}}(\mathtt{s}) = \mathbb{E}^{\pi}\left[\sum_{k=\mathtt{t}}^{\mathtt{T}} R_k \middle| \mathtt{s}_{\mathtt{t}} = \mathtt{s}\right]$$

Observe that $\mathcal{V}^{\pi}_0(\mathtt{s}_0) = \mathcal{V}^{\pi}(\mathtt{s}_0)$.

Lemma 5.5 (Fixed Policy Value Iteration). *The values $\mathcal{V}^{\pi}_{\mathtt{t}}(\mathtt{s})$ may be computed by ~~the~~ backward recursion:*

$$\mathcal{V}^{\pi}_{\mathtt{t}}(\mathtt{s}) = \left\{ \mathtt{r}_{\mathtt{t}}(\mathtt{s}, \mathtt{a}) + \sum_{\mathtt{s}' \in \mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s}, \mathtt{a})\, \mathcal{V}^{\pi}_{\mathtt{t}+1}(\mathtt{s}') \right\}_{a=\pi_{\mathtt{t}}(\mathtt{s})}, \quad \forall \mathtt{s} \in \mathcal{S}_{\mathtt{t}}$$

for $\mathtt{t} = \mathtt{T} - 1, \ldots, 0$, starting with $\mathcal{V}^{\pi}_{\mathtt{T}}(\mathtt{s}) = \mathtt{r}_{\mathtt{T}}(\mathtt{s})$.

Proof. Observe that:

$$\begin{aligned}
\mathcal{V}_{\mathtt{t}}^{\pi}(\mathtt{s}) &= \mathbb{E}^{\pi}\left[R_{\mathtt{t}} + \sum_{k=\mathtt{t}+1}^{\mathtt{T}} R_k \middle| \mathtt{s}_{\mathtt{t}} = \mathtt{s}, \mathtt{a}_{\mathtt{t}} = \pi_{\mathtt{t}}(\mathtt{s})\right] \\
&= \mathbb{E}^{\pi}\left[R_{\mathtt{t}} + \mathbb{E}^{\pi}\left[\sum_{k=\mathtt{t}+1}^{\mathtt{T}} R_k \middle| \mathtt{s}_{\mathtt{t}+1}\right] \middle| \mathtt{s}_{\mathtt{t}} = \mathtt{s}, \mathtt{a}_{\mathtt{t}} = \pi_{\mathtt{t}}(\mathtt{s})\right] \\
&= \mathbb{E}^{\pi}\left[\mathtt{r}_{\mathtt{t}}(\mathtt{s}_{\mathtt{t}}, \mathtt{a}_{\mathtt{t}}) + \mathcal{V}_{\mathtt{t}+1}^{\pi}(\mathtt{s}_{\mathtt{t}+1}) | \mathtt{s}_{\mathtt{t}} = \mathtt{s}, \mathtt{a}_{\mathtt{t}} = \pi_{\mathtt{t}}(\mathtt{s})\right] \\
&= \mathtt{r}_{\mathtt{t}}(\mathtt{s}, \pi_{\mathtt{t}}(\mathtt{s})) + \sum_{\mathtt{s}' \in \mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s}, \pi_{\mathtt{t}}(\mathtt{s}))\, \mathcal{V}_{\mathtt{t}+1}^{\pi}(\mathtt{s}').
\end{aligned}$$

The first identity is simply writing the value function explicitly, starting at state $\mathtt{s}$ at time $\mathtt{t}$ and using action $\mathtt{a} = \pi_{\mathtt{t}}(\mathtt{s})$. We split the sum to $R_{\mathtt{t}}$, the immediate reward, and the sum of other latter rewards. The second identity uses the law of total probability, we are conditioning on state $\mathtt{s}_{\mathtt{t}+1}$, and taking the expectation over it. The third identity observes that the expected value of the sum is actually the value function at $\mathtt{s}_{\mathtt{t}+1}$. The last identity writes the expectation over $\mathtt{s}_{\mathtt{t}+1}$ explicitly. This completes the proof of the lemma. □

Remark 5.6. *Note that* $\sum_{\mathtt{s}' \in \mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s}, \mathtt{a})\, \mathcal{V}_{\mathtt{t}+1}^{\pi}(\mathtt{s}') = \mathbb{E}^{\pi}[\mathcal{V}_{\mathtt{t}+1}^{\pi}(\mathtt{s}_{\mathtt{t}+1}) | \mathtt{s}_{\mathtt{t}} = \mathtt{s}, \mathtt{a}_{\mathtt{t}} = \mathtt{a}]$.

Remark 5.7. *For the more general reward function* $\tilde{\mathtt{r}}_{\mathtt{t}}(\mathtt{s}, \mathtt{a}, \mathtt{s}')$*, the recursion takes the form*

$$\mathcal{V}_{\mathtt{t}}^{\pi}(\mathtt{s}) = \left\{ \sum_{\mathtt{s}' \in \mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s}, \mathtt{a})(\tilde{\mathtt{r}}_{\mathtt{t}}(\mathtt{s}, \mathtt{a}, \mathtt{s}') + \mathcal{V}_{\mathtt{t}+1}^{\pi}(\mathtt{s}')) \right\}_{\mathtt{a}=\pi_{\mathtt{t}}(\mathtt{s})} .$$

A similar observation applies to the Dynamic Programming equations in the next section.

5.4.3 Dynamic Programming for Policy Optimization

We next define the optimal value function at each time $0 \leq \mathtt{t} \leq \mathtt{T}$:

$$\mathcal{V}_{\mathtt{t}}^{*}(\mathtt{s}) = \max_{\pi^{\mathtt{t}}} \mathbb{E}^{\pi^{\mathtt{t}}}\left[\sum_{k=\mathtt{t}}^{\mathtt{T}} R_k \middle| \mathtt{s}_{\mathtt{t}} = \mathtt{s}\right], \quad \mathtt{s} \in \mathcal{S}_{\mathtt{t}} ,$$

where the maximum is taken over "tail" policies $\pi^{\mathtt{t}} = (\pi_{\mathtt{t}}, \ldots, \pi_{\mathtt{T}-1})$ that start from time $\mathtt{t}$ and until time $\mathtt{T}-1$. Note that $\pi^{\mathtt{t}}$ is allowed to be a general policy, i.e., history-dependent and stochastic, i.e., each $\pi_k \in \pi^{\mathtt{t}}$ can depend on the history from time $\mathtt{t}$ to time k. Obviously, $\mathcal{V}_0^*(\mathtt{s}_0) = \mathcal{V}^*(\mathtt{s}_0)$.

Theorem 5.6 (Finite-horizon Dynamic Programming). *The following holds:*

1. *Backward recursion: We have that $\mathcal{V}_{\mathtt{T}}^*(\mathtt{s}) = \mathtt{r}_{\mathtt{T}}(\mathtt{s})$ for all $\mathtt{s} \in \mathcal{S}_{\mathtt{T}}$.*
   For $\mathtt{t} = \mathtt{T}-1, \ldots, 0$, we have:
$$\mathcal{V}_{\mathtt{t}}^*(\mathtt{s}) = \max_{\mathtt{a} \in \mathcal{A}_{\mathtt{t}}} \left\{ \mathtt{r}_{\mathtt{t}}(\mathtt{s}, \mathtt{a}) + \sum_{\mathtt{s}' \in \mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s}, \mathtt{a})\, \mathcal{V}_{\mathtt{t}+1}^*(\mathtt{s}') \right\}, \quad \mathtt{s} \in \mathcal{S}_{\mathtt{t}}.$$
2. *Optimal policy: Any Markov policy π^* that satisfies, for $\mathtt{t} = 0, \ldots, \mathtt{T}-1$,*
$$\pi_{\mathtt{t}}^*(\mathtt{s}) \in \arg\max_{\mathtt{a} \in \mathcal{A}_{\mathtt{t}}} \left\{ \mathtt{r}_{\mathtt{t}}(\mathtt{s}, \mathtt{a}) + \sum_{\mathtt{s}' \in \mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s}, \mathtt{a})\, \mathcal{V}_{\mathtt{t}+1}^*(\mathtt{s}') \right\}, \quad \forall \mathtt{s} \in \mathcal{S}_{\mathtt{t}},$$
   is an optimal control policy. Furthermore, π^* maximizes $\mathcal{V}^{\pi}(\mathtt{s}_0)$ simultaneously for every initial state $\mathtt{s}_0 \in \mathcal{S}_0$.

Note that Theorem 5.6 specifies an optimal control policy which is a deterministic Markov policy. Further, the theorem prescribes an algorithm for computing the optimal value function and policy.

Algorithm 7 Finite-Horizon Dynamic Programming

Initialize:
For all $\mathtt{s} \in \mathcal{S}_{\mathtt{T}}$, set:
$$V_{\mathtt{T}}(\mathtt{s}) = \mathtt{r}_{\mathtt{T}}(\mathtt{s})$$
For $\mathtt{t} = \mathtt{T}-1, \mathtt{T}-2, \ldots, 0$
  For all $\mathtt{s} \in \mathcal{S}_{\mathtt{t}}$
    Set:
$$V_{\mathtt{t}}(\mathtt{s}) = \max_{\mathtt{a} \in \mathcal{A}_{\mathtt{t}}} \left\{ \mathtt{r}_{\mathtt{t}}(\mathtt{s}, \mathtt{a}) + \sum_{\mathtt{s}' \in \mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s}, \mathtt{a})\, V_{\mathtt{t}+1}(\mathtt{s}') \right\}$$
    Store the action $\pi_{\mathtt{t}}^*(\mathtt{s})$ that achieves the maximum.

Theorem 5.6 claims that the values computed in Algorithm 7 satisfy $V_{\mathtt{t}}(\mathtt{s}) = \mathcal{V}_{\mathtt{t}}^*(\mathtt{s})$ for all $\mathtt{t} \in 0, \ldots, \mathtt{T}$, and that the deterministic policy stored by the algorithm is optimal. We now prove the theorem.

Proof. **Part (i):**

We use induction to show that the stated backward recursion in Algorithm 7 indeed yields the optimal value function $\mathcal{V}^*_{\mathtt{t}}$. The idea is simple, but some care is needed with the notation since we consider general policies, and not just Markov policies.

For the base of the induction we start with $\mathtt{t} = \mathtt{T}$. The equality $\mathcal{V}^*_{\mathtt{T}}(\mathtt{s}) = \mathtt{r}_{\mathtt{T}}(\mathtt{s})$ follows directly from the definition of $\mathcal{V}^*_{\mathtt{T}}$. Therefore, we have $V_{\mathtt{T}} = \mathcal{V}^*_{\mathtt{T}}$.

We proceed by backward induction. Suppose that $V_{\mathtt{t}+1}(\mathtt{s})$ is the optimal value function for time $\mathtt{t}+1$, i.e., $V_{\mathtt{t}+1}(\mathtt{s}) = \mathcal{V}^*_{\mathtt{t}+1}(\mathtt{s})$. We need to show that $V_{\mathtt{t}}(\mathtt{s}) = \mathcal{V}^*_{\mathtt{t}}(\mathtt{s})$, where

$$V_{\mathtt{t}}(\mathtt{s}) \triangleq \max_{\mathtt{a}\in\mathcal{A}_{\mathtt{t}}} \left\{ \mathtt{r}_{\mathtt{t}}(\mathtt{s},\mathtt{a}) + \sum_{\mathtt{s}'\in\mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a})\, V_{\mathtt{t}+1}(\mathtt{s}') \right\}.$$

We will first establish that $\mathcal{V}^*_{\mathtt{t}}(\mathtt{s}) \geq V_{\mathtt{t}}(\mathtt{s})$, and then that $\mathcal{V}^*_{\mathtt{t}}(\mathtt{s}) \leq V_{\mathtt{t}}(\mathtt{s})$.

(a) We first show that $\mathcal{V}^*_{\mathtt{t}}(\mathtt{s}) \geq V_{\mathtt{t}}(\mathtt{s})$. For that purpose, it is enough to find a policy $\pi^{\mathtt{t}}$ so that $\mathcal{V}^{\pi^{\mathtt{t}}}_{\mathtt{t}}(\mathtt{s}) = V_{\mathtt{t}}(\mathtt{s})$, since $\mathcal{V}^*_{\mathtt{t}}(\mathtt{s}) \geq \mathcal{V}^{\pi}_{\mathtt{t}}(\mathtt{s})$ for any policy π.

Fix $\mathtt{s} \in \mathcal{S}_{\mathtt{t}}$, and define $\pi^{\mathtt{t}}$ as follows: Choose $\mathtt{a}_{\mathtt{t}} = \bar{\mathtt{a}}$, where

$$\bar{\mathtt{a}} \in \arg\max_{\mathtt{a}\in\mathcal{A}_{\mathtt{t}}} \left\{ \mathtt{r}_{\mathtt{t}}(\mathtt{s},\mathtt{a}) + \sum_{\mathtt{s}'\in\mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a})\, V_{\mathtt{t}+1}(\mathtt{s}') \right\},$$

and then, after observing $\mathtt{s}_{\mathtt{t}+1} = \mathtt{s}'$, proceed with the optimal tail policy $\pi^{\mathtt{t}+1}(\mathtt{s}')$ that obtains $\mathcal{V}^{\pi^{\mathtt{t}+1}(\mathtt{s}')}_{\mathtt{t}+1}(\mathtt{s}') = V_{\mathtt{t}+1}(\mathtt{s}')$. Proceeding similarly to the proof of Lemma 5.5 (value iteration for a fixed policy), we obtain:

$$\mathcal{V}^{\pi^{\mathtt{t}}}_{\mathtt{t}}(\mathtt{s}) = \mathtt{r}_{\mathtt{t}}(\mathtt{s},\bar{\mathtt{a}}) + \sum_{\mathtt{s}'\in\mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\bar{\mathtt{a}})\, \mathcal{V}^{\pi^{\mathtt{t}+1}(\mathtt{s}')}_{\mathtt{t}+1}(\mathtt{s}') \tag{5.1}$$

$$= \mathtt{r}_{\mathtt{t}}(\mathtt{s},\bar{\mathtt{a}}) + \sum_{\mathtt{s}'\in\mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\bar{\mathtt{a}})\, V_{\mathtt{t}+1}(\mathtt{s}') = V_{\mathtt{t}}(\mathtt{s}), \tag{5.2}$$

as was required.

(b) To establish $\mathcal{V}^*_{\mathtt{t}}(\mathtt{s}) \leq V_{\mathtt{t}}(\mathtt{s})$, it is enough to show that $\mathcal{V}^{\pi^{\mathtt{t}}}_{\mathtt{t}}(\mathtt{s}) \leq V_{\mathtt{t}}(\mathtt{s})$ for any (general, randomized) "tail" policy $\pi^{\mathtt{t}}$.

Fix $\mathtt{s} \in \mathcal{S}_{\mathtt{t}}$. Consider then some tail policy $\pi^{\mathtt{t}} = (\pi_{\mathtt{t}}, \ldots \pi_{\mathtt{T}-1})$. Note that this means that $\mathtt{a}_k \sim \pi_k(\mathtt{a}|\mathtt{h}_{\mathtt{t}:k})$, where $\mathtt{h}_{\mathtt{t}:k} = (\mathtt{s}_{\mathtt{t}}, \mathtt{a}_{\mathtt{t}}, \mathtt{s}_{\mathtt{t}+1}, \mathtt{a}_{\mathtt{t}+1}, \ldots, \mathtt{s}_k)$. For each state-action pair $\mathtt{s} \in \mathcal{S}_{\mathtt{t}}$ and $\mathtt{a} \in \mathcal{A}_{\mathtt{t}}$, let $(\pi^{\mathtt{t}}|\mathtt{s},\mathtt{a})$ denote the tail policy $\pi^{\mathtt{t}+1}$ from time $\mathtt{t}+1$ onward which is obtained from $\pi^{\mathtt{t}}$ given that $\mathtt{s}_{\mathtt{t}} = \mathtt{s}$, $\mathtt{a}_{\mathtt{t}} = \mathtt{a}$. As before, by

value iteration for a fixed policy,

$$\mathcal{V}_{\mathtt{t}}^{\pi^{\mathtt{t}}}(\mathtt{s}) = \sum_{\mathtt{a}\in\mathcal{A}_{\mathtt{t}}} \pi_{\mathtt{t}}(\mathtt{a}|\mathtt{s}) \left\{ \mathtt{r}_{\mathtt{t}}(\mathtt{s},\mathtt{a}) + \sum_{\mathtt{s}'\in\mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a})\, \mathcal{V}_{\mathtt{t}+1}^{(\pi^{\mathtt{t}}|\mathtt{s},\mathtt{a})}(\mathtt{s}') \right\}.$$

But since $V_{\mathtt{t}+1}$ is optimal,

$$\begin{aligned}\mathcal{V}_{\mathtt{t}}^{\pi^{\mathtt{t}}}(\mathtt{s}) &\le \sum_{\mathtt{a}\in\mathcal{A}_{\mathtt{t}}} \pi_{\mathtt{t}}(\mathtt{a}|\mathtt{s}) \left\{ \mathtt{r}_{\mathtt{t}}(\mathtt{s},\mathtt{a}) + \sum_{\mathtt{s}'\in\mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a})\, V_{\mathtt{t}+1}(\mathtt{s}') \right\} \\ &\le \max_{\mathtt{a}\in\mathcal{A}_{\mathtt{t}}} \left\{ \mathtt{r}_{\mathtt{t}}(\mathtt{s},\mathtt{a}) + \sum_{\mathtt{s}'\in\mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a})\, V_{\mathtt{t}+1}(\mathtt{s}') \right\} = V_{\mathtt{t}}(\mathtt{s}),\end{aligned}$$

which is the required inequality in (b).

Part (ii) The main point is to show that it is sufficient that the optimal policy would be Markov (rather than history dependent) and deterministic (rather than stochastic).

We will only sketch the proof. Let π^* be the (Markov) policy defined in part 2 of Theorem 5.6. Our goal is to show that the value function of π^* coincides with that of the optimal policy, which we showed is equal to $V_{\mathtt{t}}$ that we computed. Once we show that, we prove that π^* is optimal.

Consider the value iteration (Lemma 5.5). The updates for $\mathcal{V}_{\mathtt{t}}^{\pi^*}$ in the value iteration, given the action selection of π^*, are identical to those of $V_{\mathtt{t}}$. This implies that $\mathcal{V}_{\mathtt{t}}^{\pi^*} = V_{\mathtt{t}}$ (formally, by induction on $\mathtt{t}$). Since $V_{\mathtt{t}}$ is the optimal value function, it implies that π^* is an optimal policy. □

5.4.4 The $\mathcal{Q}$ Function

Let

$$\mathcal{Q}_{\mathtt{t}}^*(\mathtt{s},\mathtt{a}) \triangleq \mathtt{r}_{\mathtt{t}}(\mathtt{s},\mathtt{a}) + \sum_{\mathtt{s}'\in\mathcal{S}_{\mathtt{t}+1}} \mathtt{p}_{\mathtt{t}}(\mathtt{s}'|\mathtt{s},\mathtt{a})\, \mathcal{V}_{\mathtt{t}+1}^*(\mathtt{s}').$$

This is known as the optimal state-action value function, or simply as the $\mathcal{Q}$-*function*. $\mathcal{Q}_{\mathtt{t}}^*(\mathtt{s},\mathtt{a})$ is the expected return from stage $\mathtt{t}$ onward, if we choose $\mathtt{a}_{\mathtt{t}} = \mathtt{a}$ and then proceed optimally.

Theorem 5.6 can now be succinctly expressed as

$$\mathcal{V}_{\mathtt{t}}^*(\mathtt{s}) = \max_{\mathtt{a}\in\mathcal{A}_{\mathtt{t}}} \mathcal{Q}_{\mathtt{t}}^*(\mathtt{s},\mathtt{a}),$$

and

$$\pi^*_{\mathtt{t}}(\mathtt{s}) \in \arg\max_{\mathtt{a} \in \mathcal{A}_{\mathtt{t}}} \mathcal{Q}^*_{\mathtt{t}}(\mathtt{s}, \mathtt{a}).$$

The $\mathcal{Q}$ function provides the basis for the $\mathcal{Q}$-learning algorithm, which is one of the basic Reinforcement Learning algorithms, and would be discussed in Chapter 11.